YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 6 OF 10 · 40 MINUTES

Ice cube rescue 冰块保卫战

A thermal insulator slows heat transfer: a coat on a snowman slows heat moving from the warm air into the snow, so it melts more slowly.

- I can plan and carry out a fair test to compare thermal insulators.
- I can explain, using my results, why a coat slows down melting (or cooling).

thermal insulator 隔热材料 · insulate 隔热 · heat transfer 热传递 · melt 熔化 · meltwater 融化的水 · temperature 温度 · thermometer 温度计 · material 材料 · concept cartoon 概念漫画

THE LESSON

Do Now	4	A: wrap 3 bagged cubes, bag A bare, note start time. B: wrap cups B to D, A bare. Goals.
Hook	3	Snowman cartoon vote: 'I agree with ___ because ___.' B: teacher pours 150 ml warm water.
Learn	7	Heat moves warm to cold; a coat slows it. Fair test, cup A control; safety. B: start readings.
Do part 1	14	Hands off cups. Variables table, prediction, Stretch, Challenge. B: end readings at 15 min.
Think 1 (B: Do continues)	4	A: groups choose a question, draft planning frame page 1. B: finish temperature drops, ranks.
Measure (B: frame page 1)	4	A: gloves on, unwrap, measure each bag's meltwater; never cut. B: planning frame page 1.
Think and Reflect	4	Reporters: best and worst material; 'Who was right?' (Owen); exit ticket; frames into folders.

Hands-on: A: bagged ice in 3 materials, 1 bare control; meltwater after about 29 min. B: 50 °C water, drop over 15 min.

Collect or assess: Worksheet write-up (Rubric strands 2 to 4); planning frame page 1; exit ticket: best insulator, number and unit, reason.

SAFETY

- Wet floor: cups on trays, mop spills at once. No eating or drinking; wash hands.
- Plan A: cold skin: gloves for ice held over a minute; ice stays in sealed bags.
- Plan B: only teacher pours water (about 50 °C, max 55 °C); never mercury thermometers

WATCH OUT

- Start the ice (A) or cup jackets (B) in the Do Now, trays out first. A: never cut the measuring slot.
- Misconceptions: 'coats make heat', 'cold comes in'. Heat moves, cold does not; shiny does not mean best.

Language: Frame: 'I agree with ___ because ___.' Chinese talk welcome; Chinese answers count. 'Insulator' here means heat.

Next week: Frames stay in folders. Now: strip trial, buy untreated sawdust. A: book freezer; B: flour, salt, drying shelf

PREP FOR ONE CLASS

Plan A: before the lesson about 80 min (60 could be done by someone else); after about 28 min (some times estimated)

Plan B: before the lesson about 75 min (55 could be done by someone else); after about 30 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book freezer or canteen ice (about 28 cubes); check 2 ice trays, cool box — Plan A · 5 min · teacher
- [ ] Reserve 7 thermometers (never mercury), 6 stopwatches, kettle, 2 jugs — Plan B · 10 min · technician

THE WEEK BEFORE

- [ ] Gather kit, 6 folders; laminate and put up A3 cartoon; label 24 cups A to D — 25 min · technician
- [ ] Send print job for your plan only (designs rebuilt after the language pass) — 5 min · reprographics
- [ ] Gather 28 zip-lock bags, 12 pairs of gloves, 6 measuring cylinders — Plan A · 5 min · technician
- [ ] Mark 150 ml line on 24 cups; check 7 thermometers agree within 1 °C — Plan B · 10 min · technician

THE DAY BEFORE

- [ ] Set project groups (tracker col F); label 6 folders, add frames; buddies — 10 min · teacher
- [ ] Freeze 2 trays of ice (about 28 cubes); cut 15 x 15 cm squares, pack trays — Plan A · 15 min · technician
- [ ] Cut 25 x 10 cm jackets and 10 x 10 cm slit lids; pack 6 trays — Plan B · 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Bag, seal 28 ice cubes; trays on tables, folders on desk; cool box at door — Plan A · 15 min · technician
- [ ] Trays out, folders on desk; boil kettle; Do Now: jug to about 50 °C, 3.6 l — Plan B · 10 min · teacher

AFTER THE LESSON

- [ ] Clean-up by roles (Plan B: teacher first counts the 7 thermometers, 2 min) — 3 min · students
- [ ] Dry and store reusable materials; mop if needed; restock the 6 trays — 10 min · technician
- [ ] Keep frames in folders for Week 7; read exit tickets; note leaders, support — 15 min · teacher

PRINT

- Worksheet, Plan A or B, x24 (3 sheets each, double-sided)
- Week 7 planning frame, Plan A or B, x8 (3 sheets each, double-sided)
- Concept cartoon x1 (A3, single-sided, then laminated)
- Teacher copy: kit card for your plan (A5, single-sided)

GET OR MAKE (NEW THIS LESSON)

- Trays x6, group folders x6, A3 laminating pouch x1
- Paper towels x36 sheets; mop and bucket (or a roll)
- Scissors and a ruler (teacher prep)
- Foil, cotton wool sheet, bubble wrap, newspaper, fabric
- Cups labelled A to D x24 (Plan B: 150 ml line marked)
- Plan A: 18 pre-cut 15 x 15 cm squares (6 sets of 3)
- Plan A: plastic cups x30, elastic bands x30
- Plan A: ice cubes x28, each sealed in a zip-lock bag
- Plan A: ice-cube trays x2, cool box x1, freezer drawer
- Plan A: disposable gloves x12 pairs
- Plan A: measuring cylinders x6 (25 ml or 10 ml)
- Plan B: 36 pre-cut pieces (18 jackets, 18 slit lids)
- Plan B: identical cups (about 250 ml) x25, bands x36
- Plan B: thermometers x7 (spirit or digital, no mercury)
- Plan B: kettle, 2 jugs (3.6 litres at about 50 °C)
- Plan B: measuring cylinder x1 (to mark the 150 ml line)

Have ready from earlier weeks: Class box with 24 logbooks, Role cards x6 sets (with buddy card), Sentence frame card sets x6, Classroom language cards x24, Science Dictionaries x24, Pencils x24; marker pen x1, Plan B: stopwatches x6, Plan A: stopwatches x6 (optional)

Chinese: draft, needs native-speaker check